Mock Interview: Chubby

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Quiz ?

Statement

How does Chubby ensure fault tolerance and reliability of data?

Answer

Fault tolerance in Chubby is achieved through data replication and maintaining a transaction log for the master.For recovery in case of failure, all database transactions are stored in a transaction log. As this transaction log can become very large over time, every few hours, the master of each Chubby cell writes a snapshot of its database to a GFS server in a different building for fault tolerance.

Hide Answer

Quiz ?

Statement

How does Chubby ensure high availability in case of master failure?

Answer

When a master fails, Chubby uses PAXOS to elect a new master. The new master loads the system state from the database and the transaction log/snapshot before serving requests.

Hide Answer

Quiz ?

Statement

How does Chubby ensure file data integrity?

Answer

Chubby uses a 64-bit checksum to detect if a file replica is different.

Hide Answer

Quiz ?

Statement

What happens when a Chubby replica fails permanently?

Answer

If a replica fails and does not recover for a few hours, a simple replacement system allocates a new server from a free pool of free machines and starts the Chubby binary on it. It then updates the DNS tables, replacing the IP address of the failed replica with that of the new one. The current master polls the DNS periodically and eventually notices the change. It then updates the list of the cell's members in the cell's database. This list is kept consistent across all the members via the normal replication protocol. In the meantime, the new replica obtains a recent copy of the database from a combination of backups stored on file servers and updates from active replicas. Once the new replica has processed a request that the current master is waiting to commit, the replica is permitted to vote in the elections for a new master.

Hide Answer

Quiz ?

Statement

How do clients find the master using Chubby?

Answer

All Chubby replicas are stored in DNS. Client contacts DNS to know the listed Chubby replicas. Client then calls any server directly in the Chubby cell via RPC. If that replica is not the master, it returns the address of the current master. Once the master is located, the client maintains a session with it and sends all requests to it until it indicates that it is not the master anymore or stops responding.

Hide Answer

Quiz ?

Statement

During cache invalidation, while the master is waiting for acknowledgments from clients, can other clients still read the file?

Answer

During the time the master is waiting for the acknowledgments from clients, the file is treated as 'uncachable'; this means that the clients can still read the file but will not cache it. This approach ensures that reads always get processed without any delay. This is useful because reads outnumber writes. To handle a client who bombards Chubby to read an 'uncached' file, an alternate design could be to block all read requests during invalidations, or a hybrid scheme can be implemented combining the two above-mentioned approaches.

Hide Answer

Quiz ?

Statement

Chubby stores its database snapshot in a GFS cluster. How does Chubby avoid cyclic dependency if that GFS cluster depends upon Chubby for electing its master?

Answer

The master of each Chubby cell writes a snapshot of its database to a GFS server in a different building. The use of a separate building ensures that the backup will survive building damage, and that the backups introduce no cyclic dependencies in the system. A GFS cell in the same building potentially might rely on the Chubby cell for electing its master.

Hide Answer

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